Adaptive Role Selection for Zero-Shot Reasoning

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Abstract

Large language models (LLMs) exhibit promising reasoning capabilities, but their zero-shot performance often suffers from rigid task-agnostic prompting. While prior work demonstrates that manually crafted role-specific prompts enhance performance, this approach lacks scalability. We propose Adaptive Role-Play Prompting, where the LLM autonomously selects contextually optimal roles for each task, eliminating manual role engineering. Experiments on arithmetic and commonsense reasoning datasets show our method outperforms Zero-Shot-CoT and matches/exceeds Role-Play Prompting on strategic tasks (70.4% on StrategyQA), highlighting its efficacy in dynamic role adaptation. This work advances scalable zero-shot reasoning by bridging the flexibility of role-specific prompting with the automation of zero-shot methods.

1. Introduction

Large language models (LLMs) have demonstrated significant progress in reasoning and comprehension tasks, but their zero-shot reasoning abilities remain challenging. Prior work has shown that assigning specific roles to prompts can enhance LLM performance (Shanahan et al., 2023; Kong et al., 2024), but manual creation of these roles is labor-intensive.

This paper proposes an innovative solution: utilizing the LLM to automatically select the most appropriate roles for each task, optimizing time and resources. Our approach, termed Adaptive Role-Play Prompting, empowers the LLM to analyze the task and choose a role that it believes will be most effective in solving it.

Through comparative experiments, we aim to demonstrate the feasibility and superiority of automated role generation in terms of effectiveness and adaptability compared to traditional Zero-Shot Reasoning techniques. We evaluate our method on datasets covering arithmetic and commonsense reasoning, comparing its performance against Zero-Shot (Kojima et al., 2023), Zero-Shot Chain-of-Thought (Zero-Shot-CoT) (Kojima et al., 2023), Few-Shot Chain-of-Thought (Few-Shot-CoT) (Wei et al., 2023), and Role-Play Prompting (Shanahan et al., 2023; Kong et al., 2024).

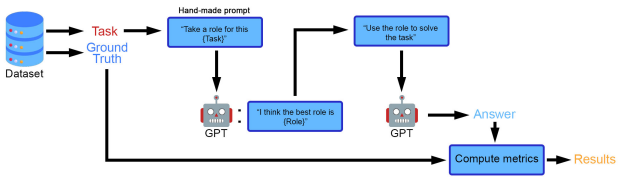

Figure 1: This block diagram illustrates the workflow of the proposed method. The dataset initially provides a task and corresponding ground truth. A hand-crafted prompt is then employed to present the task to a GPT model, which suggests an appropriate role. Subsequently, a second prompt is used to instruct the same GPT model to apply the suggested role to solve the task, thereby generating a response. Finally, metrics are computed by comparing the generated response with the ground truth, enabling an evaluation of the response's quality.

This approach has the potential to significantly improve zero-shot reasoning capabilities and reduce the need for manual intervention, leading to greater scalability and adaptability of LLMs in diverse applications.

2. Method: Adaptive Role-Playing

The core idea of Adaptive Role-Play Prompting involves prompting the LLM to select an appropriate assistant role based on the given task without actually solving it yet. In response to this initial prompt, the LLM provides a detailed description of the role and confirms its adoption. Subsequently, a new user input is added, presenting the model with the task to be solved, along with an instruction to use the chosen role for solving it. Finally, the language model delivers the final response, solving the task based on the adopted role.

The process can be broken down into four steps:

1. **Role Selection Prompt:** The initial prompt instructs the language model to select a suitable assistant role for the task at hand.

2. **Role Description Response:** The model responds by

describing in detail the role it will assume.

3. **Task Resolution Prompt:** A follow-up prompt presents the actual task to the model, instructing it to use the previously selected role to provide a solution.

4. **Final Response:** The model processes the task within the context of the chosen role and generates a response.

Figure 1 illustrates the overall workflow, and Figure 2 shows the conversational structure.

Adaptive Role-Play

```
User: Choose the best role for the
following question or task: [TASK]
You should answer with a detailed
description of the role you will take,
not with the answer to the question
or task itself.".

Assistant: Based on the task
that you provided me, I think
the best role is...[DETAILED
DESCRIPTION OF ROLE]. Now I am an
excellent....[ROLE]... and I will
take that role the best I can.

User: Now using that role,
answer/complete the following
question/task: [TASK]

Assistant: [ANSWER]
```

Figure 2: Illustration of the conversation between the user and the assistant for the Adaptive Role-Play methodology.

3. Experiments

3.1. Tasks and Datasets

To evaluate our approach, we selected datasets covering arithmetic and commonsense reasoning:

1. **Arithmetic Reasoning:**
   - **AQUA-RAT** (Ling et al., 2017): This dataset tests arithmetic reasoning and problem-solving skills through complex word problems.
2. **Commonsense Reasoning:**
   - **CSQA** (Talmor et al., 2019): This dataset focuses on assessing the model's ability to apply commonsense knowledge to answer questions.
   - **StrategyQA** (Geva et al., 2021): This dataset presents questions requiring strategic thinking and planning.

Due to computational constraints, we utilized 50% of the test dataset (randomly sampled) for StrategyQA and CSQA evaluations.

3.2. Experimental Setup

We used the `gpt-3.5-turbo` model accessed via the OpenAI API with a temperature setting of 0 for all experiments to ensure reproducibility. Our approach was compared against several established baselines: Zero-Shot prompting (Kojima et al., 2023), Zero-Shot Chain-of-Thought (Zero-Shot-CoT) (Kojima et al., 2023), Few-Shot Chain-of-Thought (Few-Shot-CoT) (Wei et al., 2023), and Role-Play Prompting (Shanahan et al., 2023; Kong et al., 2024). For Role-Play Prompting, we used manually crafted roles known to be effective for each task type (e.g., "Math Professor" for AQUA-RAT, "Expert Strategist" for StrategyQA).

4. Results

Table 1 summarizes the comparative accuracy results across the evaluated datasets. On the arithmetic reasoning task (AQUA-RAT), while manually curated Role-Play Prompting yields the highest accuracy (63.8%), our proposed Adaptive Role-Play method achieves a competitive 59.5%. Notably, this performance surpasses standard Zero-Shot (53.5%) and Zero-Shot-CoT (53.9%) baselines and achieves parity with Few-Shot-CoT (59.4%), demonstrating considerable efficacy without requiring predefined roles or few-shot examples. In the realm of commonsense reasoning, Adaptive Role-Play exhibits varied performance.

While achieving a respectable 74.0% on CSQA, slightly trailing Few-Shot-CoT (76.3%) and Role-Play Prompting (77.2%), it significantly outperforms all other methods on StrategyQA, attaining 70.4% accuracy. This marked advantage on StrategyQA underscores the potential benefit of dynamic role selection for tasks demanding nuanced or strategic reasoning. The observed variance suggests that the utility of adaptive role selection may be context-dependent, offering substantial gains in specific reasoning paradigms like StrategyQA.

We acknowledge that evaluations on CSQA and StrategyQA utilized 50% subsamples due to computational constraints, introducing potential statistical variance; however, the consistent improvement on StrategyQA provides indicative evidence of the method's robust adaptation capabilities. Validation on full datasets remains crucial future work.

5. Conclusion

This paper introduces Adaptive Role-Play Prompting, a novel approach to enhance the zero-shot reasoning capa-

Table 1: Accuracy comparison of Adaptive Role-Play with baselines on each dataset. Metrics marked with * indicate that 50% of the testing dataset was used (randomly sampled). Best performance per dataset is in **bold**.

METHOD	AQUA	CSQA	STRATEGYQA
ZERO-SHOT	53.5	74.5*	66.0*
ZERO-SHOT-COT	53.9	68.8*	65.8*
FEW-SHOT-COT	59.4	76.3*	67.4*
ROLE-PLAY PROMPTING	**63.8**	**77.2***	67.0*
ADAPTIVE ROLE-PLAY	59.5	74.0*	**70.4***

bilities of LLMs by automating the role selection process. By empowering the LLM to autonomously determine and adopt the most suitable role for a given task, our method improves performance, particularly on tasks requiring strategic or nuanced reasoning like StrategyQA, while eliminating the need for laborious manual role engineering.

The experimental results demonstrate the effectiveness of this adaptive approach, achieving performance competitive with or superior to strong baselines, including manually crafted Role-Play Prompting in some cases. This work represents a step towards more scalable and adaptable LLMs, highlighting the potential for self-optimization within prompting strategies. Future research could explore applying this method to a broader range of tasks, investigating the quality and diversity of roles chosen by the LLM, and refining the meta-prompting mechanism for role selection.

Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning, specifically in improving the reasoning capabilities of Large Language Models through automated prompting techniques. While enhancing LLM reasoning can lead to beneficial applications in areas like education, research assistance, and complex problem-solving, potential negative societal consequences could arise from misuse of more capable reasoning models (e.g., generating sophisticated misinformation, automating harmful tasks). Furthermore, the reliance on large models like GPT-3.5-turbo carries environmental costs associated with training and inference. Our work focuses on making existing models more effective in a zero-shot setting, potentially reducing the need for extensive fine-tuning or larger models for specific tasks, which could mitigate some computational burden. We encourage responsible development and deployment of the techniques presented.

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